

Project: Banking Transaction Management System

-- CREATE DATABASE

```
CREATE DATABASE BankingDB;
```

```
USE BankingDB;
```

-- 1. BRANCHES

```
CREATE TABLE Branches
```

```
(
    BranchID INT AUTO_INCREMENT PRIMARY KEY,
    BranchName VARCHAR(100) NOT NULL,
    BranchCode VARCHAR(20) NOT NULL UNIQUE,
    IFSCCode VARCHAR(20) NOT NULL UNIQUE,
    Address VARCHAR(200),
    City VARCHAR(50),
    State VARCHAR(50),
    Pincode VARCHAR(10),
    PhoneNumber VARCHAR(15),
    BranchManager VARCHAR(100),
    CreatedDate TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

-- 2. CUSTOMERS

```
CREATE TABLE Customers
```

```
(
    CustomerID INT AUTO_INCREMENT PRIMARY KEY,
```

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```
FirstName VARCHAR(50) NOT NULL,
LastName VARCHAR(50),
Gender ENUM('M','F','O'),
DateOfBirth DATE,
MobileNumber VARCHAR(15) UNIQUE,
Email VARCHAR(100) UNIQUE,
AadhaarNumber VARCHAR(20) UNIQUE,
PANNumber VARCHAR(15) UNIQUE,
Address VARCHAR(200),
City VARCHAR(50),
State VARCHAR(50),
Pincode VARCHAR(10),
CustomerSince DATE DEFAULT (CURRENT_DATE),
Status ENUM('Active','Inactive','Blocked') DEFAULT 'Active'
);
```

```
-- =====
```

-- 3. ACCOUNTS

```
-- =====
```

```
CREATE TABLE Accounts
(
    AccountID INT AUTO_INCREMENT PRIMARY KEY,
    AccountNumber VARCHAR(20) NOT NULL UNIQUE,
    CustomerID INT NOT NULL,
    BranchID INT NOT NULL,
    AccountType ENUM('Savings','Current','Salary','Fixed Deposit'),
    Balance DECIMAL(18,2) DEFAULT 0.00,
    OpenDate DATE DEFAULT (CURRENT_DATE),
    AccountStatus ENUM('Active','Inactive','Closed') DEFAULT 'Active',

    CONSTRAINT FK_Account_Customer
    FOREIGN KEY (CustomerID)
    REFERENCES Customers(CustomerID),
```

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```
CONSTRAINT FK_Account_Branch
FOREIGN KEY (BranchID)
REFERENCES Branches(BranchID)
);

-- =====
-- 4. TRANSACTIONS
-- =====

CREATE TABLE Transactions
(
    TransactionID INT AUTO_INCREMENT PRIMARY KEY,
    AccountID INT NOT NULL,
    TransactionDate DATETIME DEFAULT CURRENT_TIMESTAMP,
    TransactionType ENUM('Deposit','Withdrawal','Transfer'),
    TransactionAmount DECIMAL(18,2) NOT NULL,
    BalanceAfterTransaction DECIMAL(18,2),
    TransactionMode ENUM('Cash','ATM','UPI','NEFT','RTGS','IMPS'),
    Remarks VARCHAR(200),

    CONSTRAINT FK_Transaction_Account
    FOREIGN KEY (AccountID)
    REFERENCES Accounts(AccountID)
);

-- =====
-- 5. LOANS
-- =====

CREATE TABLE Loans
(
    LoanID INT AUTO_INCREMENT PRIMARY KEY,
    CustomerID INT NOT NULL,
    BranchID INT NOT NULL,
    LoanType ENUM('Home','Car','Personal','Education','Business'),
```

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```
LoanAmount DECIMAL(18,2),
InterestRate DECIMAL(5,2),
LoanStartDate DATE,
LoanEndDate DATE,
EMIAmount DECIMAL(18,2),
LoanStatus ENUM('Active','Closed','Pending'),
```

```
CONSTRAINT FK_Loan_Customer
FOREIGN KEY (CustomerID)
REFERENCES Customers(CustomerID),
```

```
CONSTRAINT FK_Loan_Branch
FOREIGN KEY (BranchID)
REFERENCES Branches(BranchID)
```

```
);
```

```
-- =====
```

-- 6. CARDS

```
-- =====
```

```
CREATE TABLE Cards
(
    CardID INT AUTO_INCREMENT PRIMARY KEY,
    AccountID INT NOT NULL,
    CardNumber VARCHAR(20) UNIQUE,
    CardType ENUM('Debit','Credit'),
    ExpiryDate DATE,
    CVV CHAR(3),
    CardStatus ENUM('Active','Blocked','Expired'),

    CONSTRAINT FK_Card_Account
    FOREIGN KEY (AccountID)
    REFERENCES Accounts(AccountID)
);
```

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-- 7. EMPLOYEES

```
CREATE TABLE Employees
(
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,
    BranchID INT NOT NULL,
    EmployeeName VARCHAR(100) NOT NULL,
    Designation VARCHAR(50),
    Department VARCHAR(50),
    Salary DECIMAL(12,2),
    MobileNumber VARCHAR(15),
    Email VARCHAR(100) UNIQUE,
    HireDate DATE,
    EmployeeStatus ENUM('Active','Inactive') DEFAULT 'Active',

    CONSTRAINT FK_Employee_Branch
    FOREIGN KEY (BranchID)
    REFERENCES Branches(BranchID)
);
```

-- 1. BRANCHES

```
INSERT INTO Branches
(BranchName, BranchCode, IFSCCode, Address, City, State, Pincode, PhoneNumber,
BranchManager)
VALUES
('Chennai Main','CHN001','SBIN0001001','Anna Salai','Chennai','Tamil
Nadu','600002','04422334455','Ramesh Kumar'),
('Coimbatore','CBE001','SBIN0001002','Avinashi Road','Coimbatore','Tamil
Nadu','641018','04222334455','Priya Sharma'),
('Madurai','MDU001','SBIN0001003','KK Nagar','Madurai','Tamil
Nadu','625020','04522334455','Arun Prakash');
```

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-- 2. CUSTOMERS

INSERT INTO Customers

(FirstName,LastName,Gender,DateOfBirth,MobileNumber,Email,AadhaarNumber,PANNumber,Address,City,State,Pincode)

VALUES

('Rahul','Sharma','M','1992-05-15','9876543210','rahul@gmail.com','123412341234','ABCDE1234F','Velachery','Chennai','Tamil Nadu','600042'),

('Priya','Kannan','F','1995-10-20','9876543211','priya@gmail.com','234523452345','BCDEF2345G','Anna Nagar','Chennai','Tamil Nadu','600040'),

('Suresh','Kumar','M','1990-01-08','9876543212','suresh@gmail.com','345634563456','CDEFG3456H','Race Course','Coimbatore','Tamil Nadu','641018');

-- 3. ACCOUNTS

INSERT INTO Accounts

(AccountNumber,CustomerID,BranchID,AccountType,Balance)

VALUES

('1000000000001',1,1,'Savings',250000.00),

('1000000000002',2,1,'Savings',180000.00),

('1000000000003',3,2,'Current',550000.00);

-- 4. TRANSACTIONS

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```
-- =====  
  
INSERT INTO Transactions
```

```
(AccountID,TransactionType,TransactionAmount,BalanceAfterTransaction,TransactionMode  
,Remarks)
```

```
VALUES
```

```
(1,'Deposit',50000,250000,'Cash','Cash Deposit'),
```

```
(2,'Withdrawal',20000,180000,'ATM','ATM Withdrawal'),
```

```
(3,'Transfer',100000,550000,'NEFT','Online Transfer');
```

```
-- =====  
-- 5. LOANS  
-- =====
```

```
INSERT INTO Loans
```

```
(CustomerID,BranchID,LoanType,LoanAmount,InterestRate,LoanStartDate,LoanEndDate,E  
MIAmount,LoanStatus)
```

```
VALUES
```

```
(1,1,'Home',2500000,8.50,'2024-01-10','2044-01-10',21695,'Active'),
```

```
(2,1,'Car',800000,9.25,'2025-02-15','2030-02-15',16642,'Active'),
```

```
(3,2,'Personal',500000,11.50,'2024-06-01','2029-06-01',10988,'Pending');
```

```
-- =====  
-- 6. CARDS  
-- =====
```

```
INSERT INTO Cards
```

```
(AccountID,CardNumber,CardType,ExpiryDate,CVV,CardStatus)
```

```
VALUES
```

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```
(1,'5432100011112222','Debit','2030-12-31','321','Active'),
```

```
(2,'5432100011113333','Debit','2031-08-31','456','Active'),
```

```
(3,'5432100011114444','Credit','2029-05-31','789','Blocked');
```

-- 7. EMPLOYEES

INSERT INTO Employees

(BranchID,EmployeeName,Designation,Department,Salary,MobileNumber,Email,HireDate)

VALUES

```
(1,'Vignesh Kumar','Branch  
Manager','Operations',95000,'9000011111','vignesh@bank.com','2020-05-10'),
```

```
(1,'Meena Raj','Cashier','Cash',45000,'9000011112','meena@bank.com','2022-03-15'),
```

```
(2,'Arun Kumar','Relationship  
Manager','Sales',60000,'9000011113','arun@bank.com','2021-08-20');
```

SELECT * FROM Branches;

BranchID	BranchName	BranchCode	IFSCCode	Address	City	State	Pincode	PhoneNumber	BranchManager	CreatedDate
1	Chennai Main	CHN001	SBIN0001001	Anna Salai	Chennai	Tamil Nadu	600002	04422334455	Ramesh Kumar	2026-07-14 04:28:48
2	Coimbatore	CBE001	SBIN0001002	Avinashi Road	Coimbatore	Tamil Nadu	641018	04222334455	Priya Sharma	2026-07-14 04:28:48
3	Madurai	MDU001	SBIN0001003	KK Nagar	Madurai	Tamil Nadu	625020	04522334455	Arun Prakash	2026-07-14 04:28:48
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Customers;

CustomerID	FirstName	LastName	Gender	DateOfBirth	MobileNumber	Email	AadhaarNumber	PANNumber	Address	City	State	Pincode	CustomerSince	Status
1	Rahul	Sharma	M	1992-05-15	9876543210	rahul@gmail.com	123412341234	ABCDE1234F	Velachery	Chennai	Tamil Nadu	600042	2026-07-14	Active
2	Priya	Kannan	F	1995-10-20	9876543211	priya@gmail.com	234523452345	BCDEF2345G	Anna Nagar	Chennai	Tamil Nadu	600040	2026-07-14	Active
3	Suresh	Kumar	M	1990-01-08	9876543212	suresh@gmail.com	345634563456	CDEFG3456H	Race Course	Coimbatore	Tamil Nadu	641018	2026-07-14	Active
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Accounts;

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Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

FA

	AccountID	AccountNumber	CustomerID	BranchID	AccountType	Balance	OpenDate	AccountStatus
▶	1	1000000000001	1	1	Savings	250000.00	2026-07-14	Active
	2	1000000000002	2	1	Savings	180000.00	2026-07-14	Active
	3	1000000000003	3	2	Current	550000.00	2026-07-14	Active
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Transactions;

Result Grid		Filter Rows:		Edit:		Export/Import:		Wrap Cell Content:	
	TransactionID	AccountID	TransactionDate	TransactionType	TransactionAmount	BalanceAfterTransaction	TransactionMode	Remarks	
▶	1	1	2026-07-14 04:29:31	Deposit	50000.00	250000.00	Cash	Cash Deposit	
	2	2	2026-07-14 04:29:31	Withdrawal	20000.00	180000.00	ATM	ATM Withdrawal	
	3	3	2026-07-14 04:29:31	Transfer	100000.00	550000.00	NEFT	Online Transfer	
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	

SELECT * FROM Loans;

Result Grid											Filter Rows:	Edit:	Export/Import:	Wrap Cell Content:
	LoanID	CustomerID	BranchID	LoanType	LoanAmount	InterestRate	LoanStartDate	LoanEndDate	EMIAmount	LoanStatus				
▶	1	1	1	Home	2500000.00	8.50	2024-01-10	2044-01-10	21695.00	Active				
	2	2	1	Car	800000.00	9.25	2025-02-15	2030-02-15	16642.00	Active				
	3	3	2	Personal	500000.00	11.50	2024-06-01	2029-06-01	10988.00	Pending				
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL				

SELECT * FROM Cards;

Result Grid

Filter Rows:

Edit:

Export/Import:

	CardID	AccountID	CardNumber	CardType	ExpiryDate	CVV	CardStatus
▶	1	1	5432100011112222	Debit	2030-12-31	321	Active
	2	2	5432100011113333	Debit	2031-08-31	456	Active
	3	3	5432100011114444	Credit	2029-05-31	789	Blocked
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL

SELECT * FROM Employees;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

EmployeeID	BranchID	EmployeeName	Designation	Department	Salary	MobileNumber	Email	HireDate	EmployeeStatus
1	1	Vignesh Kumar	Branch Manager	Operations	95000.00	9000011111	vignesh@bank.com	2020-05-10	Active
2	1	Meena Raj	Cashier	Cash	45000.00	9000011112	meena@bank.com	2022-03-15	Active
3	2	Arun Kumar	Relationship Manager	Sales	60000.00	9000011113	arun@bank.com	2021-08-20	Active
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Customer Account Details

SELECT

c.CustomerID,

CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,

a.AccountNumber,

a.AccountType,

a.Balance

FROM Customers c

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INNER JOIN Accounts a

ON c.CustomerID = a.CustomerID;

RESULT

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	CustomerID	CustomerName	AccountNumber	AccountType	Balance
▶	1	Rahul Sharma	100000000001	Savings	250000.00
	2	Priya Kannan	100000000002	Savings	180000.00
	3	Suresh Kumar	100000000003	Current	550000.00

##Customer + Branch Details

SELECT

c.CustomerID,

CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,

b.BranchName,

a.AccountNumber,

a.Balance

FROM Customers c

JOIN Accounts a

ON c.CustomerID = a.CustomerID

JOIN Branches b

ON a.BranchID = b.BranchID;

RESULT

Result Grid



Filter Rows:

Export:


Wrap Cell Content:

	CustomerID	CustomerName	BranchName	AccountNumber	Balance
▶	1	Rahul Sharma	Chennai Main	1000000000001	250000.00
	2	Priya Kannan	Chennai Main	1000000000002	180000.00
	3	Suresh Kumar	Coimbatore	1000000000003	550000.00

Total Balance by Branch

SELECT

b.BranchName,

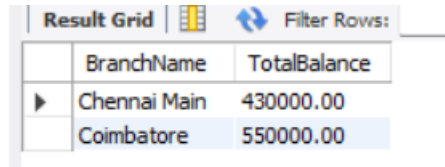
SUM(a.Balance) AS TotalBalance

FROM Branches b

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```
JOIN Accounts a
ON b.BranchID = a.BranchID
GROUP BY b.BranchName;
```

RESULTS



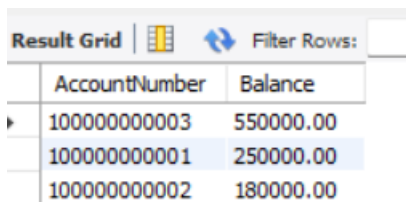
A screenshot of a database application's 'Result Grid'. It features a toolbar with icons for 'Result Grid', a grid icon, a refresh icon, and a 'Filter Rows:' dropdown. The grid contains two columns: 'BranchName' and 'TotalBalance'. There are two rows: 'Chennai Main' with a balance of 430000.00, and 'Coimbatore' with a balance of 550000.00.

	BranchName	TotalBalance
▶	Chennai Main	430000.00
	Coimbatore	550000.00

##Top 5 Customers by Balance

```
SELECT
    AccountNumber,
    Balance
FROM Accounts
ORDER BY Balance DESC
LIMIT 5;
```

RESULT



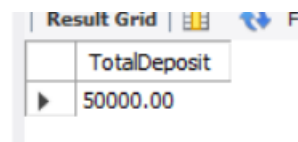
A screenshot of a database application's 'Result Grid'. It features a toolbar with icons for 'Result Grid', a grid icon, a refresh icon, and a 'Filter Rows:' dropdown. The grid contains two columns: 'AccountNumber' and 'Balance'. There are three rows: '1000000000003' with a balance of 550000.00, '1000000000001' with a balance of 250000.00, and '1000000000002' with a balance of 180000.00.

	AccountNumber	Balance
▶	1000000000003	550000.00
	1000000000001	250000.00
	1000000000002	180000.00

##Total Deposit Amount

```
SELECT
SUM(TransactionAmount) AS TotalDeposit
FROM Transactions
WHERE TransactionType='Deposit';
```

RESULT



A screenshot of a database application's 'Result Grid'. It features a toolbar with icons for 'Result Grid', a grid icon, a refresh icon, and a 'Filter Rows:' dropdown. The grid contains one column: 'TotalDeposit'. There is one row with a value of 50000.00.

	TotalDeposit
▶	50000.00

##Total Withdrawal Amount

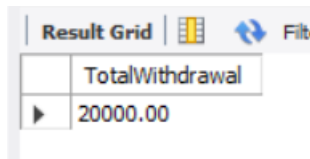
```
SELECT
SUM(TransactionAmount) AS TotalWithdrawal
```

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FROM Transactions

WHERE TransactionType='Withdrawal';

RESULT



TotalWithdrawal
20000.00

##Latest Transaction

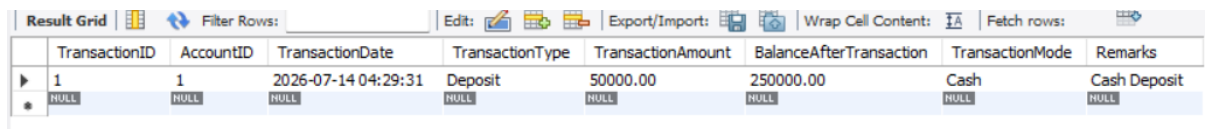
SELECT *

FROM Transactions

ORDER BY TransactionDate DESC

LIMIT 1;

RESULT



TransactionID	AccountID	TransactionDate	TransactionType	TransactionAmount	BalanceAfterTransaction	TransactionMode	Remarks
1	1	2026-07-14 04:29:31	Deposit	50000.00	250000.00	Cash	Cash Deposit

##Monthly Transaction Report

SELECT

DATE_FORMAT(TransactionDate,'%Y-%m') AS Month,

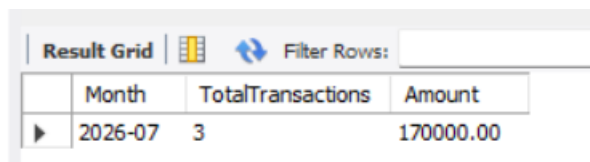
COUNT(*) AS TotalTransactions,

SUM(TransactionAmount) AS Amount

FROM Transactions

GROUP BY DATE_FORMAT(TransactionDate,'%Y-%m');

RESULT



Month	TotalTransactions	Amount
2026-07	3	170000.00

##Premium Customers

SELECT

CustomerID,

Balance,

CASE

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```
WHEN Balance >= 1000000 THEN 'Platinum'
WHEN Balance >= 500000 THEN 'Gold'
WHEN Balance >= 100000 THEN 'Silver'
ELSE 'Regular'
END AS CustomerCategory
FROM Accounts;
```

RESULT

Result Grid



Filter Rows:

	CustomerID	Balance	CustomerCategory
▶	1	250000.00	Silver
	2	180000.00	Silver
	3	550000.00	Gold

Window Function

```
SELECT
AccountID,
TransactionAmount,
ROW_NUMBER() OVER(
PARTITION BY AccountID
ORDER BY TransactionDate DESC
) AS RowNum
FROM Transactions;
```

RESULT

Result Grid



Filter Rows:

	AccountID	TransactionAmount	RowNum
▶	1	50000.00	1
	2	20000.00	1
	3	100000.00	1

Running Balance

```
SELECT
AccountID,
TransactionDate,
TransactionAmount,
SUM(TransactionAmount)
```

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```
OVER(  
PARTITION BY AccountID  
ORDER BY TransactionDate  
) AS RunningBalance  
FROM Transactions;
```

RESULT

	CustomerID	TotalBalance
▶	1	250000.00
	2	180000.00
	3	550000.00

##Common Table Expression (CTE)

```
WITH CustomerBalance AS  
(  
SELECT  
CustomerID,  
SUM(Balance) AS TotalBalance  
FROM Accounts  
GROUP BY CustomerID  
)  
SELECT *  
FROM CustomerBalance;
```

RESULT

	FirstName	LoanType	LoanAmount
▶	Rahul	Home	2500000.00
	Priya	Car	800000.00

##Active Loan Customers

```
SELECT  
c.FirstName,  
l.LoanType,  
l.LoanAmount  
FROM Customers c
```

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JOIN Loans 1

ON c.CustomerID=1.CustomerID

WHERE 1.LoanStatus='Active';

RESULT

Result Grid			Filter Rows:
	BranchID	TotalEmployees	
▶	1	2	
	2	1	

##Branch Employee Count

SELECT

BranchID,

COUNT(*) AS TotalEmployees

FROM Employees

GROUP BY BranchID;

RESULT

Result Grid				Filter Rows:
	TransactionDay	TotalTransactions	TotalAmount	
▶	2026-07-14	3	170000.00	

##Daily Transaction Summary

SELECT

DATE(TransactionDate) AS TransactionDay,

COUNT(*) AS TotalTransactions,

SUM(TransactionAmount) AS TotalAmount

FROM Transactions

GROUP BY DATE(TransactionDate);

RESULT

Result Grid				Filter Rows:
	TransactionDay	TotalTransactions	TotalAmount	
▶	2026-07-14	3	170000.00	

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##Customer Account Summary

```
SELECT
    c.CustomerID,
    CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
    a.AccountNumber,
    a.AccountType,
    a.Balance,
    b.BranchName
FROM Customers c
JOIN Accounts a
ON c.CustomerID = a.CustomerID
JOIN Branches b
ON a.BranchID = b.BranchID;
```

RESULT

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	CustomerID	CustomerName	AccountNumber	AccountType	Balance	BranchName
▶	1	Rahul Sharma	1000000000001	Savings	250000.00	Chennai Main
	2	Priya Kannan	1000000000002	Savings	180000.00	Chennai Main
	3	Suresh Kumar	1000000000003	Current	550000.00	Coimbatore

##Daily Transaction Report

```
SELECT
    DATE(TransactionDate) AS TransactionDate,
    COUNT(*) AS TotalTransactions,
    SUM(TransactionAmount) AS TotalAmount
FROM Transactions
GROUP BY DATE(TransactionDate)
ORDER BY TransactionDate DESC;
```

RESULT

Result Grid	Filter Rows:	Export:
TransactionDate	TotalTransactions	TotalAmount
2026-07-14	3	170000.00

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##Top 10 Customers by Balance

```
SELECT
    CONCAT(c.FirstName,' ',c.LastName) AS CustomerName,
    a.AccountNumber,
    a.Balance
FROM Customers c
JOIN Accounts a
ON c.CustomerID=a.CustomerID
ORDER BY a.Balance DESC
LIMIT 10;
```

RESULT

	CustomerName	AccountNumber	Balance
▶	Suresh Kumar	1000000000003	550000.00
	Rahul Sharma	1000000000001	250000.00
	Priya Kannan	1000000000002	180000.00

##Monthly Deposits

```
SELECT
    DATE_FORMAT(TransactionDate,'%Y-%m') AS Month,
    SUM(TransactionAmount) AS TotalDeposit
FROM Transactions
WHERE TransactionType='Deposit'
GROUP BY DATE_FORMAT(TransactionDate,'%Y-%m');
```

RESULTS

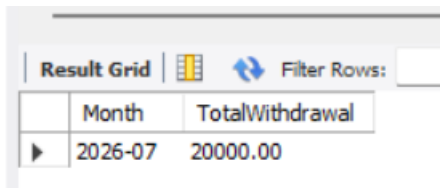
	Month	TotalDeposit
▶	2026-07	50000.00

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##Monthly Withdrawals

```
SELECT
    DATE_FORMAT(TransactionDate,%Y-%m') AS Month,
    SUM(TransactionAmount) AS TotalWithdrawal
FROM Transactions
WHERE TransactionType='Withdrawal'
GROUP BY DATE_FORMAT(TransactionDate,%Y-%m');
```

RESULT



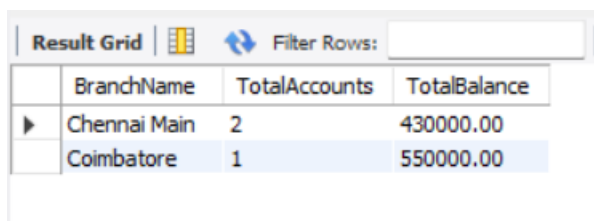
The screenshot shows a 'Result Grid' window with a 'Filter Rows' input field. The grid contains two columns: 'Month' and 'TotalWithdrawal'. There is one data row for the month '2026-07' with a total withdrawal of '20000.00'.

	Month	TotalWithdrawal
▶	2026-07	20000.00

##Branch Performance

```
SELECT
    b.BranchName,
    COUNT(a.AccountID) AS TotalAccounts,
    SUM(a.Balance) AS TotalBalance
FROM Branches b
JOIN Accounts a
ON b.BranchID=a.BranchID
GROUP BY b.BranchName;
```

RESULT



The screenshot shows a 'Result Grid' window with a 'Filter Rows' input field. The grid contains four columns: 'BranchName', 'TotalAccounts', and 'TotalBalance'. There are two data rows: 'Chennai Main' with 2 accounts and a total balance of 430000.00, and 'Coimbatore' with 1 account and a total balance of 550000.00.

	BranchName	TotalAccounts	TotalBalance
▶	Chennai Main	2	430000.00
	Coimbatore	1	550000.00

##Loan Outstanding Report

```
SELECT
    CONCAT(c.FirstName,' ',c.LastName) AS CustomerName,
    b.BranchName,
    l.LoanType,
    l.LoanAmount,
```

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```
1.EMIAmount,  
1.LoanStatus  
FROM Loans l  
JOIN Customers c  
ON l.CustomerID=c.CustomerID  
JOIN Branches b  
ON l.BranchID=b.BranchID  
WHERE l.LoanStatus='Active';
```

RESULT

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	CustomerName	BranchName	LoanType	LoanAmount	EMIAmount	LoanStatus
▶	Rahul Sharma	Chennai Main	Home	2500000.00	21695.00	Active
	Priya Kannan	Chennai Main	Car	800000.00	16642.00	Active

##Credit Card Usage

```
SELECT  
    CardType,  
    COUNT(*) AS TotalCards  
FROM Cards  
GROUP BY CardType;
```

RESULT

Result Grid	Filter Rows:
CardType	TotalCards
Debit	2
Credit	1

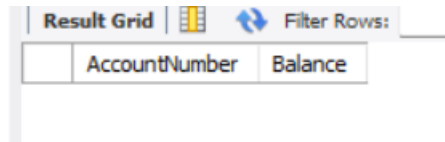
##Dormant Accounts

```
SELECT  
    a.AccountNumber,  
    a.Balance  
FROM Accounts a  
LEFT JOIN Transactions t
```

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```
ON a.AccountID=t.AccountID
GROUP BY a.AccountID
HAVING MAX(t.TransactionDate) < DATE_SUB(CURDATE(),INTERVAL 180 DAY)
OR MAX(t.TransactionDate) IS NULL;
```

RESULT

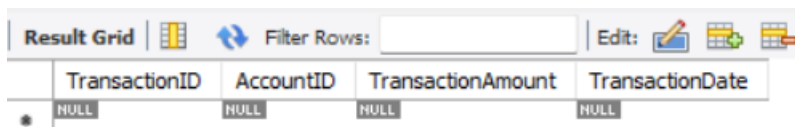


AccountNumber	Balance
---------------	---------

##Fraud Detection (High-Value Transactions)

```
SELECT
    TransactionID,
    AccountID,
    TransactionAmount,
    TransactionDate
FROM Transactions
WHERE TransactionAmount > 100000;
```

RESULT



TransactionID	AccountID	TransactionAmount	TransactionDate
NULL	NULL	NULL	NULL

##Latest Transaction per Customer

```
SELECT *
FROM
(
    SELECT
        c.CustomerID,
        CONCAT(c.FirstName, ' ',c.LastName) AS CustomerName,
        t.TransactionAmount,
        t.TransactionDate,
        ROW_NUMBER() OVER
```

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```
(  
    PARTITION BY c.CustomerID  
    ORDER BY t.TransactionDate DESC  
    ) AS rn  
FROM Customers c  
JOIN Accounts a  
ON c.CustomerID=a.CustomerID  
JOIN Transactions t  
ON a.AccountID=t.AccountID  
)x  
WHERE rn=1;
```

RESULT

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	CustomerID	CustomerName	TransactionAmount	TransactionDate	rn
▶	1	Rahul Sharma	50000.00	2026-07-14 04:29:31	1
	2	Priya Kannan	20000.00	2026-07-14 04:29:31	1
	3	Suresh Kumar	100000.00	2026-07-14 04:29:31	1

##Running Account Balance

```
SELECT  
    AccountID,  
    TransactionDate,  
    TransactionAmount,  
    SUM(TransactionAmount)  
    OVER  
    (  
        PARTITION BY AccountID  
        ORDER BY TransactionDate  
    ) AS RunningBalance  
FROM Transactions;
```

RESULT

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Result Grid Filter Rows: Export: Wrap Cell Content:				
	AccountID	TransactionDate	TransactionAmount	RunningBalance
▶	1	2026-07-14 04:29:31	50000.00	50000.00
	2	2026-07-14 04:29:31	20000.00	20000.00
	3	2026-07-14 04:29:31	100000.00	100000.00

##Top Branch by Deposits

```
SELECT
    b.BranchName,
    SUM(t.TransactionAmount) AS TotalDeposit
FROM Branches b
JOIN Accounts a
ON b.BranchID=a.BranchID
JOIN Transactions t
ON a.AccountID=t.AccountID
WHERE t.TransactionType='Deposit'
GROUP BY b.BranchName
ORDER BY TotalDeposit DESC
LIMIT 1;
```

RESULT

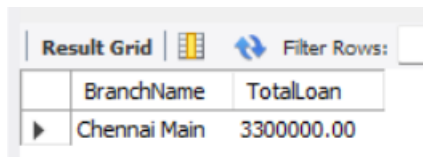
Result Grid Filter Rows:		
	BranchName	TotalDeposit
▶	Chennai Main	50000.00

##Top Branch by Loan Amount

```
SELECT
    b.BranchName,
    SUM(l.LoanAmount) AS TotalLoan
FROM Branches b
JOIN Loans l
ON b.BranchID=l.BranchID
GROUP BY b.BranchName
ORDER BY TotalLoan DESC
LIMIT 1;
```

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RESULT



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with two columns: 'BranchName' and 'TotalLoan'. The first row of data shows 'Chennai Main' and '3300000.00'. Above the table, there is a 'Filter Rows:' button and a search input field.

BranchName	TotalLoan
Chennai Main	3300000.00

##Full Banking Report (Single Query)

SELECT

c.CustomerID,
CONCAT(c.FirstName, ' ', c.LastName) AS CustomerName,
c.MobileNumber,
c.Email,

a.AccountNumber,
a.AccountType,
a.Balance,

b.BranchName,
b.City AS BranchCity,

t.TransactionDate,
t.TransactionType,
t.TransactionAmount,
t.TransactionMode,

l.LoanType,
l.LoanAmount,
l.EMIAmount,
l.LoanStatus,

ca.CardNumber,
ca.CardType,
ca.CardStatus,

e.EmployeeName,
e.Designation

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FROM Customers c

LEFT JOIN Accounts a

ON c.CustomerID = a.CustomerID

LEFT JOIN Branches b

ON a.BranchID = b.BranchID

LEFT JOIN Transactions t

ON a.AccountID = t.AccountID

LEFT JOIN Loans l

ON c.CustomerID = l.CustomerID

LEFT JOIN Cards ca

ON a.AccountID = ca.AccountID

LEFT JOIN Employees e

ON b.BranchID = e.BranchID

ORDER BY

c.CustomerID,

t.TransactionDate DESC;

RESULT

Result Grid																
Filter Rows				Export												
Wrap Cell Contents																
Result Grid	CustomerID	CustomerName	MobileNumber	Email	AccountNumber	AccountType	Balance	BranchName	BranchCity	TransactionDate	TransactionType	TransactionAmount	TransactionMode	LoanType	LoanAmount	EMIAmount
1		Rahul Sharma	9876543210	rahul@gmail.com	100000000001	Savings	250000.00	Chennai Main	Chennai	2026-07-14 04:29:31	Deposit	50000.00	Cash	Home	2500000.00	21695.00
2		Rahul Sharma	9876543210	rahul@gmail.com	100000000001	Savings	250000.00	Chennai Main	Chennai	2026-07-14 04:29:31	Deposit	50000.00	Cash	Home	2500000.00	21695.00
3		Priya Kannan	9876543211	priya@gmail.com	100000000002	Savings	180000.00	Chennai Main	Chennai	2026-07-14 04:29:31	Withdrawal	20000.00	ATM	Car	800000.00	16642.00
4		Priya Kannan	9876543211	priya@gmail.com	100000000002	Savings	180000.00	Chennai Main	Chennai	2026-07-14 04:29:31	Withdrawal	20000.00	ATM	Car	800000.00	16642.00
5		Suresh Kumar	9876543212	suresh@gmail.com	100000000003	Current	550000.00	Coimbatore	Coimbatore	2026-07-14 04:29:31	Transfer	100000.00	NEFT	Personal	500000.00	10988.00

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LoanStatus	CardNumber	CardType	CardStatus	EmployeeName	Designation
Active	5432100011112222	Debit	Active	Meena Raj	Cashier
Active	5432100011112222	Debit	Active	Vignesh Kumar	Branch Manager
Active	5432100011113333	Debit	Active	Meena Raj	Cashier
Active	5432100011113333	Debit	Active	Vignesh Kumar	Branch Manager
Pending	5432100011114444	Credit	Blocked	Arun Kumar	Relationship Manager